



CLIMATE IMPACT ON LIVESTOCK PRODUCTION IN NIGERIA (2000-2020).

Using FAO production data and climate data from NASA

Year
All

Live...
Cattle

16M

Total Production

25.30

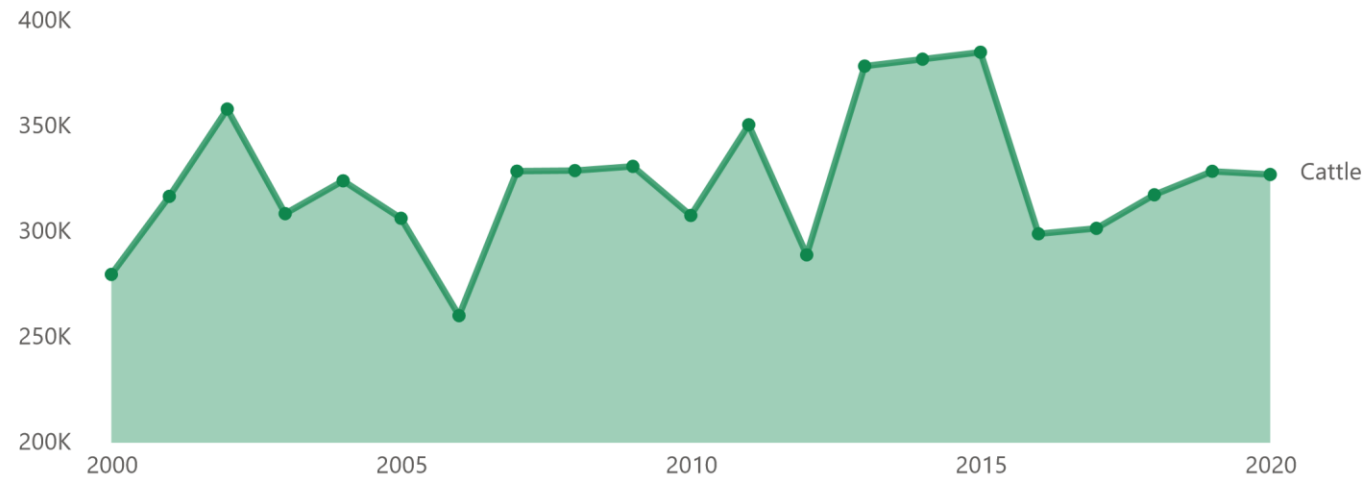
Avg Temp (°C)

2.87

Avg Rainfall (mm)

What Livestock Type Dominated Over 20 Years?

Livestock_Type ● Cattle



Which Season Had The Highest Rainfall?

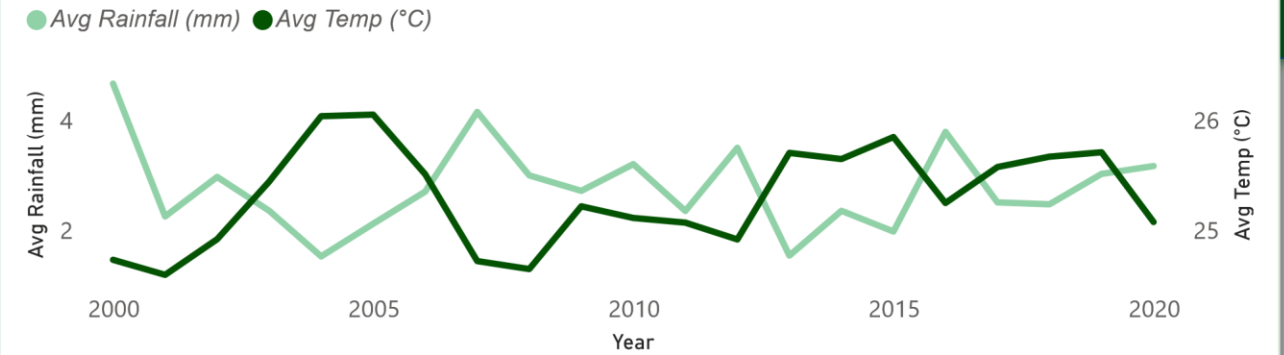


Which Season Was The Hottest?

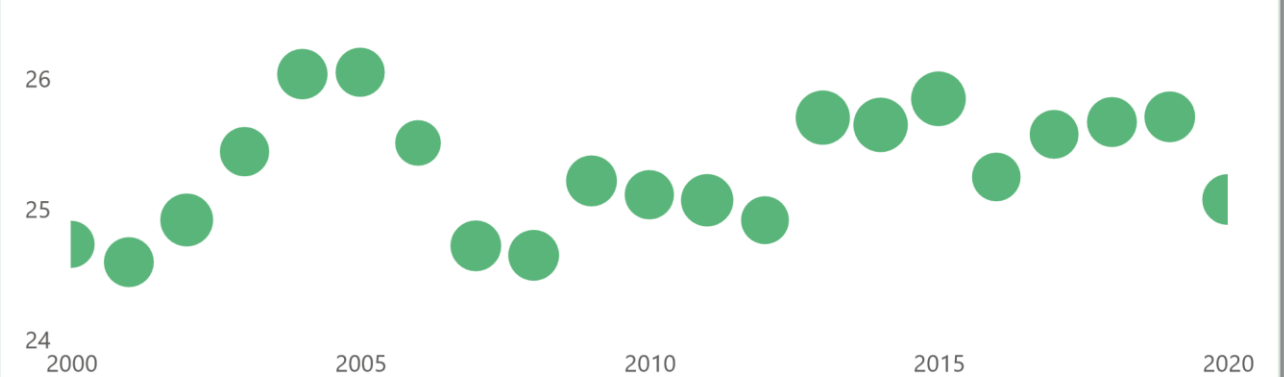


Key Finding: Despite the climate variability, livestock production in Nigeria remained stable, suggesting production is driven by non-climatic factors

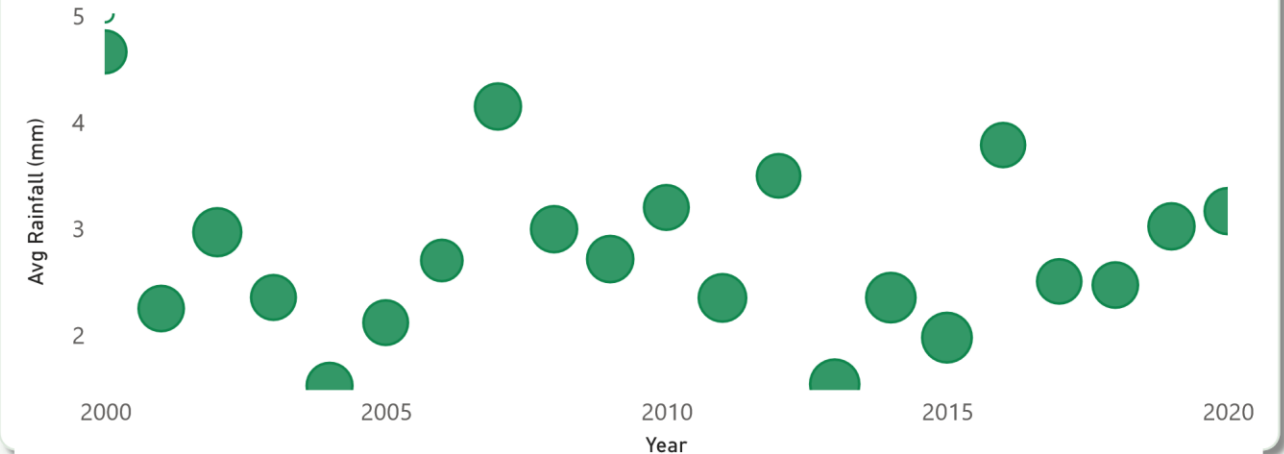
How Did Temperature And Rainfall Trend over 20 years?



Did Temperature Influence Livestock Production?



Did Rainfall Influence Livestock Production?





Key Findings & Insights:

What the data tells us about climate and livestock production in Nigeria

Production Stability;

Total livestock production in Nigeria remained relatively stable between 2000 and 2020 despite significant fluctuations in both temperature and rainfall. This suggests that livestock production is resilient to short term climate variability

Temperature Findings;

Average temperatures ranged between 24°C and 26°C over the study period. Analysis shows no strong correlation between temperature changes and livestock output, indicating that the temperature range experienced was within tolerable limits for cattle, goat and sheep production

Rainfall Findings;

Rainfall showed clear seasonal variation across the 20 year period. However this variation did not translate into significant changes in livestock production levels, suggesting that water availability may not be the primary constraint for livestock farmers in Nigeria

Livestock Type Analysis;

Cattle consistently dominated total production across all 20 years followed by Goat and Sheep. This dominance remained stable regardless of climate conditions, reflecting the cultural and economic importance of cattle in Nigerian agriculture

Conclusion;

This analysis of 20 years of FAO livestock production data and NASA climate data reveals that livestock production in Nigeria demonstrates strong resilience to climate variability. The findings suggest that non-climatic factors such as agricultural policy, veterinary services, feed availability and market access may be more significant drivers of production outcomes. Future research should explore socioeconomic variables alongside climate data to build a more complete picture of livestock productivity in Nigeria